

Linear System Fall 2026

Homework Two

Due: Friday, Oct 2

Exercise 1 Obtain the A, B, C, D matrices for a state space representation of the following systems:

(a)

$$\begin{aligned}u &= a_0 q + a_1 \dot{q} + \cdots + a_{n-1} q^{(n-1)} + q^{(n)} \\y &= \beta_0 q + \beta_1 \dot{q} + \cdots + \beta_{n-1} q^{(n-1)} + \gamma u\end{aligned}$$

where $u(t), y(t) \in \mathbb{R}$.

(b)

$$\begin{aligned}u(k) &= a_0 q(k) + a_1 q(k+1) + \cdots + a_{n-1} q(k+n-1) + q(k+n) \\y(k) &= \beta_0 q(k) + \beta_1 q(k+1) + \cdots + \beta_{n-1} q(k+n-1) + \gamma u(k)\end{aligned}$$

where $y(k), u(k) \in \mathbb{R}$.

(c) The ‘simple structure’ with outputs $y_1 = q_1$ and $y_2 = q_2$.

Exercise 2 Linearize (if possible) the following systems about *each* of their equilibrium states (if not possible, state why) and obtain the system matrix for these linearized systems.

(a)

$$\dot{x} = x^3$$

(b)

$$\dot{x} = \sqrt{|x|}$$

(c) The ‘first nonlinear system’

(d) The ‘attitude dynamics’ system.

$$\begin{aligned}I_1 \dot{\omega}_1 &= (I_2 - I_3) \omega_2 \omega_3 \\I_2 \dot{\omega}_2 &= (I_3 - I_1) \omega_3 \omega_1 \\I_3 \dot{\omega}_3 &= (I_1 - I_2) \omega_1 \omega_2\end{aligned}$$

Consider non-symmetric case ($I_1 \neq I_2 \neq I_3$).

Exercise 3 (a) Obtain all equilibrium states of the following system:

$$\begin{aligned}\dot{x}_1 &= 2x_2(1 - x_1) - x_1 \\ \dot{x}_2 &= 3x_1(1 - x_2) - x_2\end{aligned}$$

(b) Linearize the above system about the zero equilibrium state.

Exercise 4 For each of the following systems, linearize about each equilibrium solution and obtain the system A -matrix for a state space representation of these linearized systems.

(a)

$$\ddot{y} + (y^2 - 1)\dot{y} + y = 0.$$

where $y(t)$ is a scalar.

(b)

$$\ddot{y} + \dot{y} + y - y^3 = 0$$

where $y(t)$ is a scalar.

(c)

$$\begin{aligned}(M + m)\ddot{y} + ml\ddot{\theta}\cos\theta - ml\dot{\theta}^2\sin\theta + ky &= 0 \\ ml\ddot{y}\cos\theta + ml^2\ddot{\theta} + mgl\sin\theta &= 0\end{aligned}$$

where $y(t)$ and $\theta(t)$ are scalars.

(d)

$$\ddot{y} + 0.5\dot{y}|\dot{y}| + y = 0.$$

where $y(t)$ is a scalar.

Exercise 5 Obtain the transfer function (matrix) of the following system

$$\begin{aligned}\ddot{y}_1 + \ddot{y}_2 + y_1 + y_2 &= u_1 + \dot{u}_2 \\ 2\ddot{y}_1 + 3\ddot{y}_2 + y_1 - y_2 &= 0\end{aligned}$$

Exercise 6 Obtain the transfer function of the system with input u and output y described by

$$\ddot{q}_1 + 3\dot{q}_2 + \dot{q}_1 + 2q_2 = \dot{u} + 4u$$

$$\ddot{q}_1 + 4\dot{q}_2 + 3q_2 = u$$

$$y = q_1 + q_2$$